Ex $G = \mathbb{Z}$: $0, \pm 1, \pm 2, \pm 3, \ldots$

H : even integers ← a subgroup

$$H = 2\mathbb{Z}$$

multiples of 3 : $3\mathbb{Z}$ $\ldots -6, -3, 0, 3, 6, \ldots$

also a subgroup

Every subgroup of $\mathbb{Z}$ is of the form

$n\mathbb{Z}$ for $n = 0, 1, 2, 3, 4, \ldots$

Direct sums of groups:

Let G, H be groups (comm) w/ operations $+_G, +_H$

$G \oplus H$: (g, h) $g \in G, h \in H$

$(g_1, h_1) + (g_2, h_2) \equiv (g_1 + g_2, h_1 + h_2)$

$0_{G \oplus H} = (0_G, 0_H)$ identity